

Full Length Article

Boosting the selectivity of higher alcohols in upgrading of aqueous bioethanol over amphiphilic NiSn@C-MgO catalyst

Xueying Cai, Xiyin Li, Minyang Dang, Huanxin Huang, Ailin Liu, Qian Zhang, Yunhong Pi, Qingwei Meng, Xiaoping Wu^{*}, Tiejun Wang^{*}

Guangdong Provincial Key Laboratory of Plant Resources Biorefinery, School of Chemical Engineering and Light Industry, Guangdong University of Technology, Guangzhou 510006, China

ARTICLE INFO

Keywords:

Aqueous ethanol upgrading
Blending bio-fuel
Higher alcohols
Amphiphilic
NiSn@C-MgO
Carbonization temperature

ABSTRACT

Bio-ethanol produced from biomass has been regarded as a promising feedstock for producing higher alcohols (C_{4+} alcohols) through the Guerbet route. Nevertheless, the high-efficiency upgrading of bio-ethanol is driven by the current disadvantage of low selectivity to C_{6+} alcohols. Herein, we designed and synthesized an amphiphilic NiSn@C-MgO, which was composed of NiSn nanoparticles (NPs) coated with carbon shell and MgO nanoflakes support. Detailed experiments reveal that the amphiphilic NiSn@C-MgO catalyst can stabilize emulsions and facilitate the cross-coupling of ethanol and C_{4+} alcohols (generated by ethanol self-condensation) to generate C_{6+} alcohols in aqueous ethanol upgrading. Furthermore, the carbonization temperature has a great influence on the structure of NiSn@C-MgO, thus greatly affecting the catalyst performance. Consequently, the optimal NiSn@C-MgO-550 displays an excellent activity with ethanol conversion of 73.3 %, the selectivities of C_{6+} alcohols and C_{8+} alcohols in liquid products also reach up to 60.9 % and 32.5 %, respectively. This present strategy provides an efficient strategy for the direct upgrading of water-containing bio-ethanol to desired biofuel components, which may promote the fundamental researches on the production of biorefineries and green fuels.

1. Introduction

Biomass is a promising renewable resource that can be used to produce bio-based fuels or chemicals, such as bioethanol [1–3]. As increasing availability of ethanol extracted from biomass, the conversion of which into butanol [4–6] and C_{6+} alcohols (contained C_6 , i.e., hexanol) [7] has drawn widespread attention. These alcohols are widely used in the chemical industry as feedstocks and components for high value chemicals and fuels [8]. As fuel, butanol can be used as a fuel additive for gasoline due to its higher energy density and phase compatibility with gasoline [9], C_{6+} alcohols can be blended with diesel fuel [10,11], and C_{8+} alcohols are precursors of bio-jet fuel [10,12,13]. The coupling of ethanol via Guerbet reaction is a significant process to produce C_{4+} alcohols through bio-ethanol upgrading [14,15]. However, the mass transfer of substrates can easily be hindered during the process from water phase to oil–water two-phase, resulting in poor selectivity of C_{4+} alcohols. Therefore, developing efficient catalysts for ethanol upgrading to C_{4+} alcohols is of great importance [7,14,16].

Phase transfer catalyst (PTC) is well known for facilitating the

migration of a reactant in a heterogeneous system from one phase into another phase during reaction process [17,18] through changing the solubility of ions, increasing the reactivity of ions, and accelerating the reaction speed. Typical PTCs are surfactants such as quaternary ammonium salts [19,20], phosphonium salts [21,22], crown ethers [17,23] and cryptand [24], which can increase the interfacial surface area through emulsification and promote the transfer between two phases [18,25]. However, due to the difficult separation and recovery of target product [26], solid particles are gradually applied to stabilize organic aqueous emulsions. Generally, oxide nanoparticles can efficiently stabilize oil-in-water emulsions due to their hydrophilicity [27]. Low-dimensional carbon nanomaterials can also produce emulsions, but they have water-in-oil characteristics because of their hydrophobicity [28]. By integrating low-dimensional carbon nanomaterials with oxide particles to constitute hybrid nanoparticles, the hydrophilicity-hydrophobicity balance is likely to change by adjusting the composition structure. Crossley et al. [29] prepared Pd/SWNT-SiO₂ phase transfer catalyst by depositing Pd on carbon nanotube-inorganic oxide hybrid nanoparticles, showing excellent two-phase hydrogenation

^{*} Corresponding authors.

E-mail addresses: wuxp@gdut.edu.cn (X. Wu), tjwang@gdut.edu.cn (T. Wang).

<https://doi.org/10.1016/j.fuel.2022.126971>

Received 4 September 2022; Received in revised form 31 October 2022; Accepted 25 November 2022

Available online 10 December 2022

0016-2361/© 2022 Elsevier Ltd. All rights reserved.

activity for biomass refining deoxygenation and condensation catalysis in three substrate categories. Qiu et al. [30] reported Ru/LDH-CNTs by depositing Ru on CNTs and LDH, which indicated an outstanding catalytic performance for selective oxidation of benzyl alcohol to benzaldehyde with good capability for assembling and stability at the water-oil interface.

Inspired by the novel concept [29] that solid nanoparticles can simultaneously stabilize and catalyze the reaction in the two phases, we here precisely regulate the hydrophilic and lipophilic properties of NiSn composite nanoparticles via surface modifications with MgO and carbide. The prepared NiSn@C-MgO phase transfer catalyst can efficiently inhibit the deep dehydrogenation of ethanol and facilitate the oil-water two-phase mass transfer process, effectively promoting the reaction substrate to achieve secondary carbon chain growth during the self-coupling of aqueous ethanol. Excitingly, the optimal catalyst displays an excellent ethanol conversion reaching up to 73.3 %, as well as the selectivity and the yield of C₆₊ alcohols increased to 60.9 % and 36.0 %, respectively. This work provides an efficient strategy for higher alcohols production by direct upgrading of aqueous ethanol from biomass fermentation.

2. Experimental section

2.1. Catalyst preparation

2.1.1. Preparation of NiSn@C-MgO

NiSn@C-MgO catalysts were prepared by a facile sol-gel method followed by pyrolysis and carbonization strategy. 10 mmol Ni(NO₃)₂·6H₂O, 20 mmol Mg(NO₃)₂·6H₂O, 20 mmol C₆H₈O₇·H₂O and 0.5 mmol SnCl₄·5H₂O were successively dissolved in 60 mL deionized water under continuous stirring to obtain a homogeneous solution. Subsequently, the obtained mixture was gradually gelled by evaporation at 100 °C and then dried at 80 °C for 48 h in an oven. Finally, the resultant precursor was calcined in a N₂ flow of 20 mL/min at 550 °C for 3 h, with the yielded catalyst labeled as NiSn@C-MgO-550 (abbreviated as NiSn@C-MgO). The catalysts with carbonization temperature of 400 °C, 500 °C, 600 °C and 700 °C were denoted as NiSn@C-MgO-400, NiSn@C-MgO-500, NiSn@C-MgO-600 and NiSn@C-MgO-700, respectively.

2.1.2. Preparation of NiSn/MgO

NiSn/MgO was synthesized through the pyrolysis method followed by hydrogen reduction. Firstly, 10 mmol Ni(NO₃)₂·6H₂O, 20 mmol Mg(NO₃)₂·6H₂O and 0.5 mmol SnCl₄·5H₂O were successively dissolved in deionized water (60 mL) under continuous stirring to obtain a homogeneous solution at room temperature. After vigorously stirring for 2 h, the mixture was dried at 80 °C for 24 h in an oven. Finally, the resultant precursor was reduced at 550 °C under 20 mL/min H₂ flow for 3 h.

2.1.3. Preparation of NiSn@C

NiSn@C was synthesized by sol-gel method and one-step pyrolysis/carbonization strategy. Firstly, 10 mmol Ni(NO₃)₂·6H₂O, 20 mmol C₆H₈O₇·H₂O and 0.5 mmol SnCl₄·5H₂O were successively dissolved in 60 mL deionized water under continuous stirring to obtain a homogeneous solution. Subsequently, the obtained mixture was gradually gelled by evaporation at 100 °C and then dried at 80 °C for 48 h in an oven. Finally, the resultant precursor was calcined at 550 °C under 20 mL/min N₂ flow for 3 h.

2.2. Catalyst characterization

The crystalline structure of catalysts were analysed by power X-ray diffraction (XRD) on MiniFlex 600 diffractometer with Cu-K radiation. The surface properties of catalysts were determined by X-ray photoelectron spectroscopy (XPS) on a Thermo Fisher Scientific Escalab 250 Xi photoelectron spectrometer. TEM and HRTEM images of the catalysts were taken with a JEM-20100F microscope. The contact angles with the

dodecanol and water drop were observed and measured on a Data-physics OCA100 instrument. The droplet of emulsion was characterized by a polarizing microscope. More details of catalyst characterization were given in the [Supporting Information](#).

2.3. Reaction evaluation

All experiments of ethanol upgradation were performed in the 50 mL batch reactor with a stirring and the heating functions (Fig. S1: MS-50-316L, Anhui Kemi Machinery Technology Co., Ltd, Hefei, China). In a typical experiment run, 20.0 g of ethanol aqueous solution (equal mass of ethanol and deionized water), 0.5 g of NaOH and 0.5 g of catalyst were added into the reactor. Then, the reactor was sealed and filled with nitrogen to 6 MPa for leakage detection. Afterwards the reactor was flushed with hydrogen for 5 times and then maintained at 0.1 MPa hydrogen atmosphere. Finally, the temperature of reactor was heated to 250 °C with 4 °C/min before a 12 h hold, with the continuously stirring rate of 500 rpm. After the reaction, the reactor was cooled down to room temperature. The product detection and collection mainly consist of gas product and liquid product. Gas sample was collected with the air bags, and the liquid sample obtained by centrifugation was weighed and used for subsequent analysis.

The gas product was analyzed by gas chromatography (GC-9800, Kechuang, China) equipped with two thermal conductivity detector (TCDs) and a flame ionization detector (FID). Before analysis, the solid impurities and catalyst in the liquid product were removed by centrifugation and filtration, and separated into aqueous phase and organic phase. The liquid product was analyzed quantitatively and qualitatively by GC-MS (Agilent, HP-INNOWax, Thermo Fisher Scientific TRACE 1300ISQ) using an internal standard method.

The ethanol conversion, yield, selectivity and distribution of the products were calculated according to previous report [31], and the following equations were used to calculate.

$$\text{Ethanol conversion (C)} = \frac{\text{Moles of converted ethanol}}{\text{Moles of the inlet ethanol}} \times 100\% \quad (1)$$

$$\text{Selectivity (S)} = \frac{\text{Moles of carbon in the target product}}{\text{Moles of carbon in all products}} \times 100\% \quad (2)$$

$$\text{Yield (Y)} = \frac{\text{Moles of carbon in the product}}{\text{Moles of carbon in ethanol fed to the reactor}} \times 100\% \quad (3)$$

3. Results and discussion

3.1. Catalyst characterization

NiSn@C-MgO, synthesized via a facile sol-gel method followed by pyrolysis and carbonization strategy, displays amphipathy due to the co-existence of carbon shell (lipophilic) and MgO (hydrophilic) (Fig. 1). Specifically, citric acid was used as reductant, surface ligand, as well as the carbon shell precursor, while MgO was used as the hydrophilic support. Conventionally, the structure, thickness and morphology of carbon shell are greatly affected by carbonization temperature, which has a significant impact on catalyst performance, especially for amphiphilic catalysts [32–34]. X-ray diffraction (XRD) measurement was performed to study the composition of the material and the structure or morphology of the atoms or molecules. Fig. 2a shows the X-ray diffraction (XRD) of the NiSn@C-MgO catalysts prepared at different carbonization temperatures. The peaks at 42.8°, 62.2° and 78.5° in the XRD patterns are ascribed to the (200), (220) and (222) facets of MgO (JCPDS No. 74-1225). Moreover, the peaks at 44.5°, 51.8° and 76.4° are corresponded to Ni (111), (200) and (220) facets, respectively (JCPDS No. 70–1849). The XRD diffraction peaks of Ni and MgO with lower peak intensities are observed at the carbonization temperature of 400 °C. With the increase of carbonization temperature, the diffraction

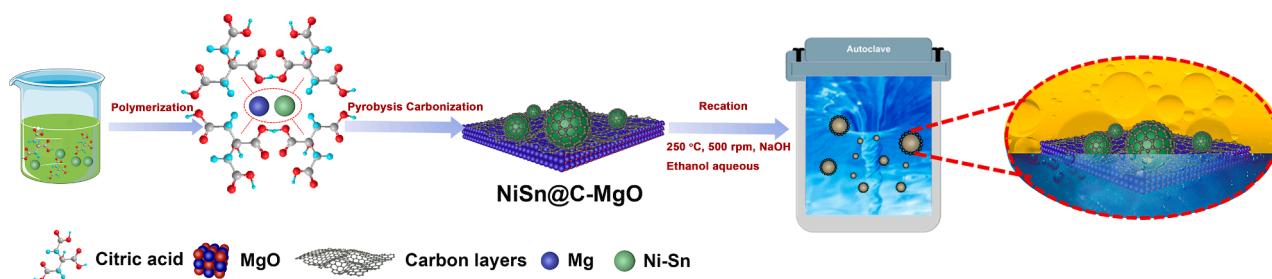


Fig. 1. Schematic illustration of the synthesis process for the NiSn@C-MgO.

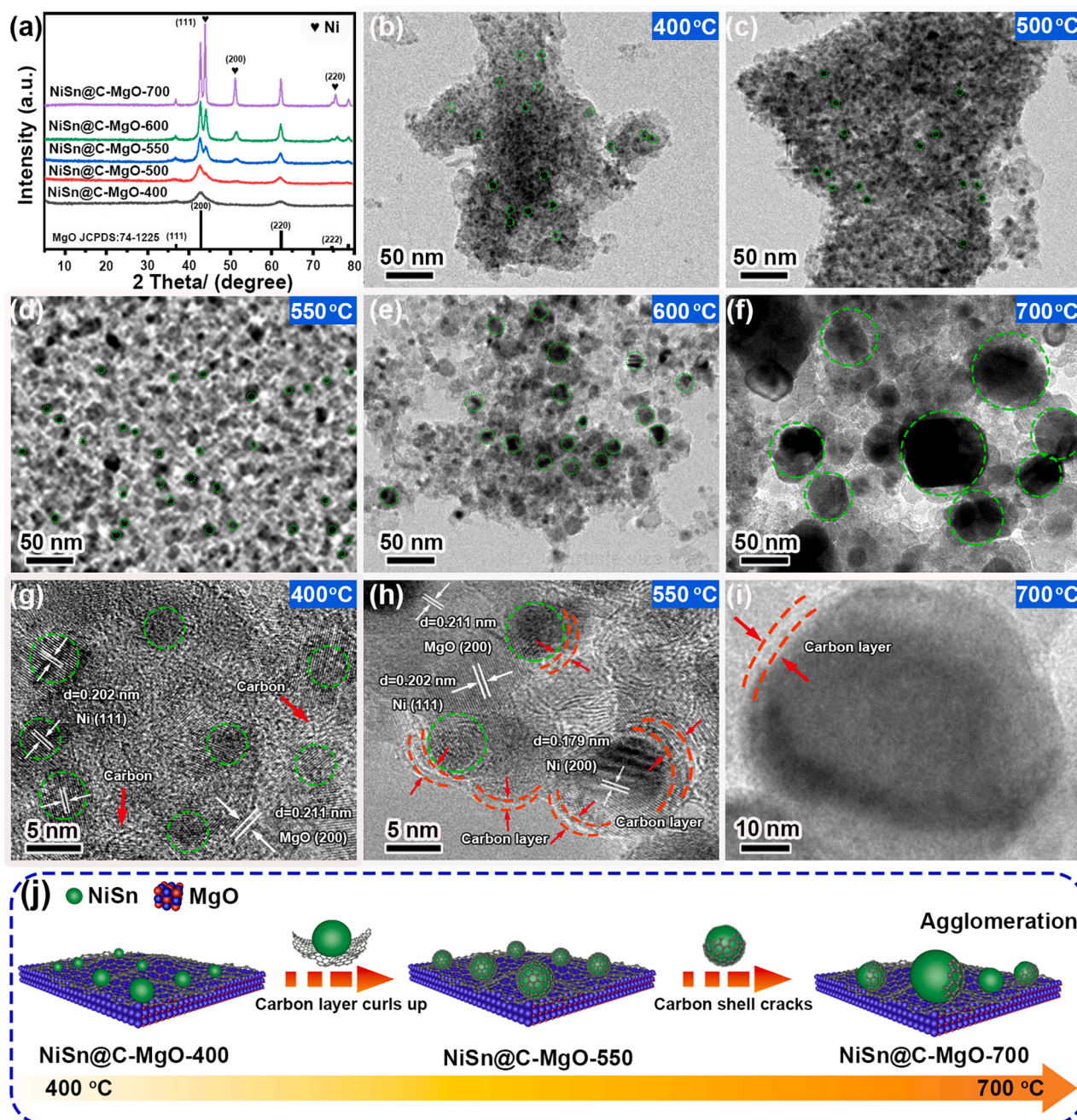


Fig. 2. A) XRD patterns of NiSn@C-MgO catalysts prepared at different carbonization temperatures. TEM images of b) NiSn@C-MgO-400, c) NiSn@C-MgO-500, d) NiSn@C-MgO-550, e) NiSn@C-MgO-600, and f) NiSn@C-MgO-700. HRTEM images of g) NiSn@C-MgO-400, h) NiSn@C-MgO-550, and i) NiSn@C-MgO-700. j) Schematic illustration for the evolution of NiSn@C-MgO catalysts with increasing carbonization temperature.

peaks of Ni become sharper and the peak width become narrower, indicating the increase in crystallite size [35,36]. In view of this, the crystallite size of the particles for NiSn/C-MgO-400, NiSn/C-MgO-500, NiSn/C-MgO-550, NiSn/C-MgO-600 and NiSn/C-MgO-700 are calculated by Scherrer equation to be 2.3, 6.0, 9.3, 18.1 and 33.2 nm, respectively. Note that there are indistinctive diffraction peaks of Sn and C in the XRD pattern, which are attributed to the low loading of Sn and the amorphous state of carbon shell, respectively.

Subsequently, the morphology, size and structure of the catalysts were characterized by transmission electron microscopy (TEM). As shown in Fig. 2b–f, NiSn nanoparticles (NPs) coated with carbon shell (green circle) are loaded on MgO support for all NiSn/C-MgO samples. Obviously, the dispersion and size of NPs are significantly impacted by carbonization temperature. In detail, the average size of particles increased from 3.4 ± 0.8 to 9.1 ± 2.1 nm slowly with carbonization temperature from 400 to 550 °C and then increased sharply to 54.7 ± 15.4 nm when the carbonization temperature reached 700 °C (Fig. 3). Furthermore, NiSn/C-MgO samples were characterized by HRTEM and the images displayed that NiSn NPs coated with carbon shell (about 1–2 nm) were loaded on MgO support (Fig. 2g–i). The lattice distance of 0.202, 0.179 and 0.211 nm are indexed as the (111), (200) facet of Ni and the (200) facet of MgO, respectively (Fig. 2g, h). For NiSn/C-MgO-400, there is no distinct carbon shell around the NiSn NPs, namely the amorphous carbon and NiSn NPs are relatively separated, which means that the NiSn nanoparticles are not successfully wrapped by the carbon shell (Fig. 2g). As the carbonization temperature increases, the amorphous carbon gradually bends and curls up, and finally wraps the NiSn NPs (Fig. 2h). However, excessive carbonization temperature will lead to the cracking of the carbon shell (Fig. 2i). Once the protection of the carbon shell is lost, NiSn NPs agglomerate and sinter seriously at a high temperature [37,38]. Therefore, the average size of NiSn NPs increase violently to 54.7 nm at carbonization temperature of 700 °C, which is consistent with the XRD results. Consequently, a scheme for the structural evolution of NiSn/C-MgO with the change of carbonization temperature is given in Fig. 2j.

The chemical states of NiSn/C-MgO samples prepared at different carbonization temperatures were further analyzed by XPS in Fig. 4, and the surface content of Ni and Sn with different chemical states are shown

in Tables S1 and S2. Fig. 4a shows the full XPS spectra of NiSn/C-MgO samples, in which the C 1s, Sn 3d, and Ni 2p peaks appeared at about 284, 486 and 854 eV, respectively. As shown in Fig. 4b, the peak at 284.8 eV corresponds to sp^2 -hybridized graphitic carbon C—C bond [39,40]. Note that the peaks at 286.1 and 288.5 eV are attributed to C—O group and C=O group in carbonyl, respectively, and they become weaken as carbonization temperature increases, implying the gradual decomposition of citric acid precursor [41]. In Ni 2p XPS spectra, the peaks at 852.8 and 853.7 eV are ascribed to Ni^0 and Ni^{2+} , respectively, indicating the partial reduction of Ni species which existed during the carbonization process. (Fig. 4c). Conventionally, more metallic oxide species are reduced to the metallic state with increasing carbonization temperature [42]. Indeed, the proportion of metallic state Ni^0 increased monotonically from 77.7 % to 83.3 % as the temperature raised from 400 to 700 °C, but the increase is not significant. The reason lies in that on the one hand, the citric acid has been mostly carbonized at 400 °C, and on the other hand, the solid solution structure of Ni(Mg)O makes it difficult for Ni^{2+} to be reduced to Ni^0 [31]. In addition, the presence of Ni^{2+} is also partly attributed to the surface oxides formed during passivation process [43]. XPS peaks of Sn 3d over NiSn/C-MgO samples with different carbonization temperatures are shown in Fig. 4d. For all samples, the Sn 3d_{5/2} spectra can be deconvoluted into three peaks at about 485.2, 486.0 and 486.6 eV, which are assigned to Sn^0 , Sn^{2+} and Sn^{4+} , respectively, demonstrating that oxidized species of Sn (Sn^{4+}) are reduced to a lower valence oxidized species of Sn (Sn^{2+}) and then to metallic Sn species (Sn^0) [44]. The XPS spectra of Ni 2p and Sn 3d for all catalysts show that oxidation state of Ni&Sn and metallic state of Ni&Sn are coexisting, thereinto, the metallic state are predominating (Table S1 and Table S2). Overall, as the carbonization temperature increases, more oxidized species are reduced to metallic species.

3.2. Catalytic performance

Firstly, the effect of carbonization temperature of NiSn/C-MgO catalysts on catalytic performance in the upgrading of aqueous ethanol was investigated. As shown in Fig. 5a, with the increase of carbonization temperature from 400 to 700 °C, the ethanol conversion displays a volcano shape, which first increase from 62.5 % to 73.3 % and

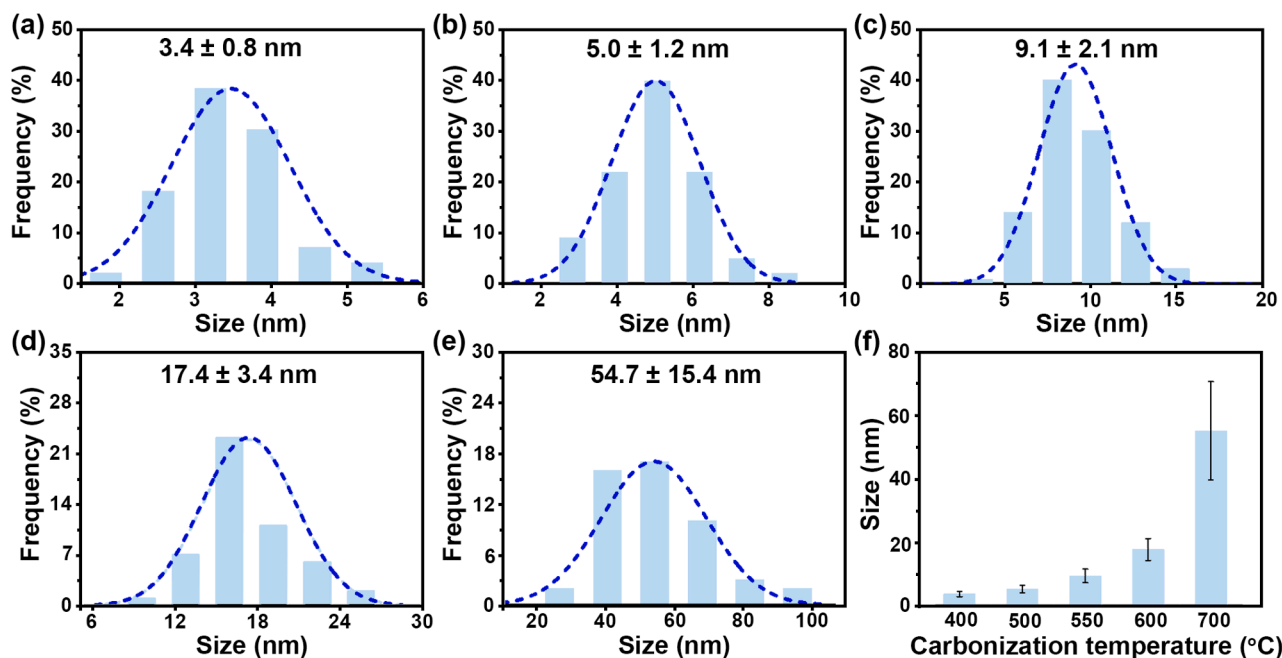


Fig. 3. Size distribution images of NiSn/C-MgO catalysts prepared at different carbonization temperatures: a) NiSn/C-MgO-400, b) NiSn/C-MgO-500, c) NiSn/C-MgO-550, d) NiSn/C-MgO-600, e) NiSn/C-MgO-700 and f) variation trend of particle sizes with carbonization temperature.

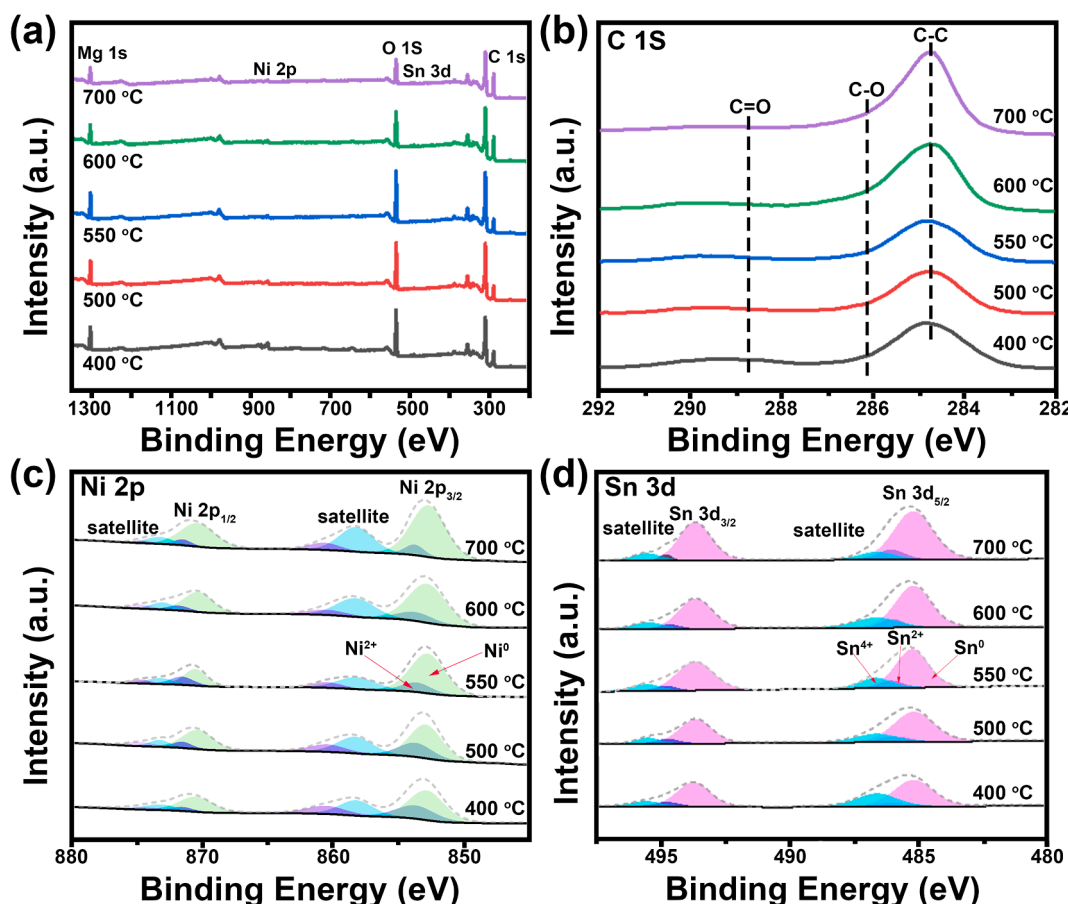


Fig. 4. A) full xps profiles, b) c 1s, c) Ni 2p, and d) Sn 3d XPS spectra of NiSn@C-MgO catalysts prepared at different carbonization temperatures.

then decrease to 60.6 %. Interestingly, NiSn@C-MgO-550 catalyst exhibited the highest ethanol conversion of 73.3 %, as well as the liquid product yield of 59.0 %. The TEM results reveal that the structure of the NiSn@C-MgO-550 catalyst is that Ni-Sn NPs coated with carbon shell supported on MgO surface; thus we consider that this special structure has excellent catalytic activity and stability for the C–C coupling of ethanol, which is due to the protection of the carbon shell that makes the Ni-Sn NPs difficult to agglomerate and sinter.

Fig. 5b shows the effect of carbonization temperature on liquid products distribution. Surprisingly, when the carbonization temperature is lower than 500 °C or higher than 600 °C, the butanol content increases significantly, while the C₆₊ alcohols obviously decreased. For example, the selectivity of butanol in liquid products reaches 51.5 %, but the selectivity of C₆₊ alcohols is only 43.2 % at carbonization temperature of 400 °C. The selectivities of C₆₊ alcohols and C₈₊ alcohols reach the maximum value of 60.9 % and 32.5 % at carbonization temperature of 550 °C, respectively. Fig. 5c and Fig. 5d show the carbon number distribution and yield of liquid products, respectively. Clearly, when the carbonization temperature is 550 °C, the yield of C₆-14 alcohols are significantly higher than that of other temperatures. These results indicate that NiSn@C-MgO-550 catalyst is more favorable for the secondary cross-coupling during the coupling of aqueous ethanol, which may be related to its special structure (Fig. 2j).

Furthermore, the effects of reaction temperature on the ethanol upgrading over NiSn@C-MgO-550 were investigated. As expected, the ethanol conversion consistently increased from 56.0 % to 76.1 % as the reaction temperature increased from 230 to 260 °C, indicating that the high reaction temperature was favorable for the conversion of ethanol (Fig. 6a). Nevertheless, higher reaction temperature also leads to intensification of aqueous phase reforming reaction, resulting in more gaseous by-products such as CH₄, CO, CO₂, and H₂. Correspondingly,

when the reaction temperature was set as 250 °C, the yield of the organic phase reached a maximum of 59.0 %, while it has a relatively high ethanol conversion of 73.3 % (Fig. 6a). More importantly, the yield of C₄₊ and C₆₊ higher alcohols also reached the maximum of 54.2 % and 36.0 %, respectively (Fig. 6b).

The reusability of NiSn@C-MgO catalyst for aqueous ethanol upgrading to C₆₊ alcohols was investigated. The catalyst was firstly separated and recovered through centrifugation and then washed with deionized water. More than 10 % mass loss in each recovery due to the amphipathicity of NiSn@C-MgO. Therefore, some fresh catalyst were added to make up the loss of catalyst in the catalyst recycling operations. As shown in Table 1, the NiSn@C-MgO catalyst exhibited excellent stability during five consecutive cycles. Ethanol conversion and C₆₊ alcohols selectivity slightly decreased from 73.3 % to 71.3 % and 60.9 % to 57.9 % after 4th cycle, respectively. The excellent stability of NiSn@C-MgO catalyst is attributed to the carbon shell as lipophilic support to protect NiSn nanoparticles from agglomeration [41].

3.3. Effect of hydrophilicity and lipophilicity on the reaction

As we mentioned above, the aqueous ethanol upgradation is a process of changing from aqueous phase to oil–water two-phase, amphiphilic NiSn@C-MgO catalyst can facilitate mass transfer of reaction substrates. In view of this, the effect of hydrophilicity/lipophilicity on aqueous ethanol upgradation were investigated over NiSn/MgO (hydrophilic), NiSn@C (lipophilic), NiSn@C-MgO (amphiphilic) and the simple physical mixing of NiSn/MgO and NiSn@C catalysts (for comparison). For the hydrophilic NiSn/MgO catalyst, the ethanol conversion and organic liquid products yield are only 50.8 % and 38.2 %, respectively (Fig. 7a). What's more, the selectivity of C₆₊ alcohols is just 42.4 % (Fig. 7b), which reveals that ethanol is difficult to produce higher

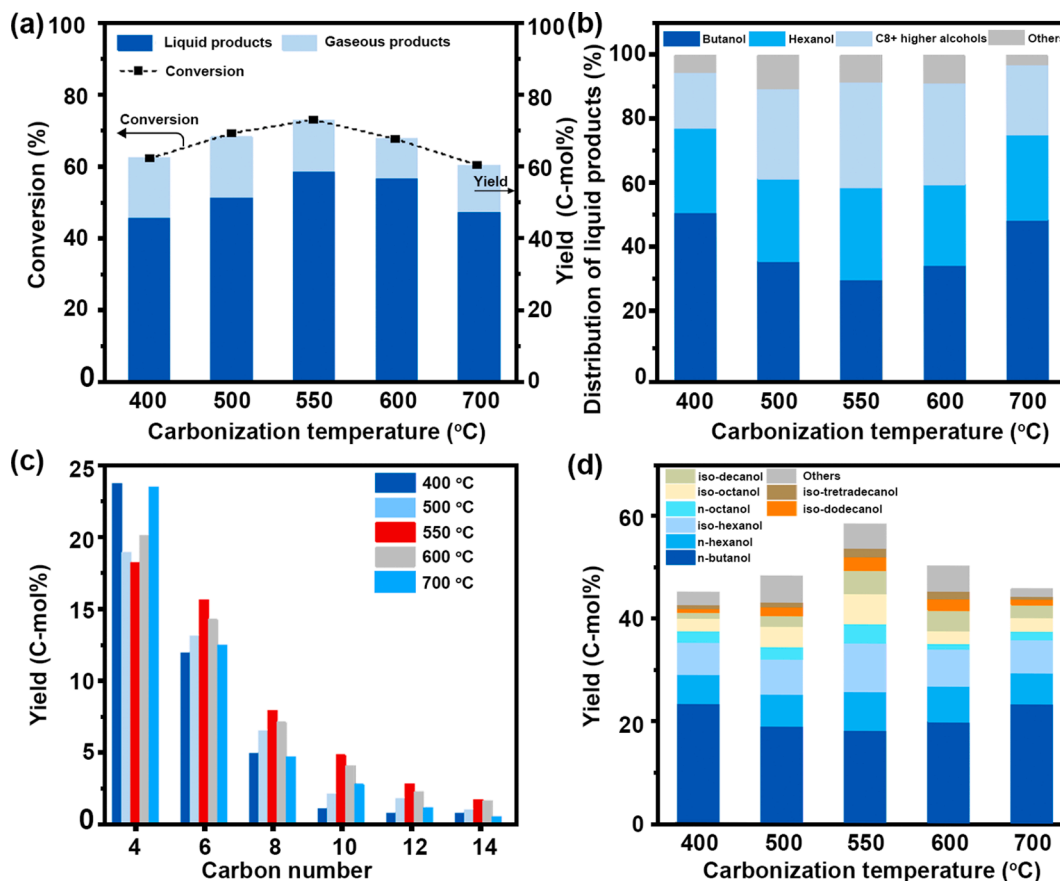


Fig. 5. A) ethanol conversion, yield of products; b) selectivity of liquid products; c) carbon number distributions of higher alcohols; d) yield of liquid products in ethanol upgradation over nsn@c-mgo catalysts prepared at different carbonization temperatures. reaction conditions: $m_{\text{NaOH}} = 0.5$ g, $m_{\text{catalyst}} = 0.5$ g, $m_{\text{ethanol}} = 10$ g, $m_{\text{water}} = 10$ g, $T = 250$ °C, and $t = 12$ h. Others: small amounts of aldehydes, trace amounts of alkanes, and other unidentified species.

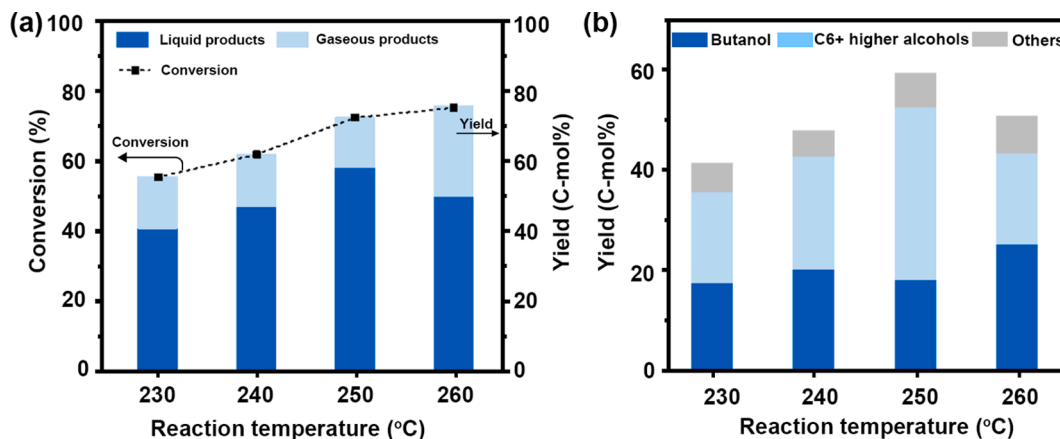


Fig. 6. Catalytic performance towards NiSn@C-MgO (different reaction temperature. a) Ethanol conversion, yield of products, and b) yields of liquid product. Reaction conditions: $m_{\text{NaOH}} = 0.5$ g, $m_{\text{catalyst}} = 0.5$ g, $m_{\text{ethanol}} = 10$ g, $m_{\text{water}} = 10$ g and $t = 12$ h.

alcohols over the hydrophilic NiSn/MgO. The ethanol conversion and liquid product selectivity for lipophilic NiSn@C catalyst are to some extent similar to that for hydrophilic NiSn/MgO (Fig. 7a–d). Strikingly, the amphiphilic NiSn@C-MgO exhibits excellent performance with ethanol conversion of 73.3 %, C_{6+} alcohols and C_{8+} alcohols reach the maximum value of 60.9 % and 32.5 %, respectively. The product distribution between water and oil phase also reveals that NiSn@C-MgO catalyst has higher selectivity for C_{6+} alcohols (Table S3). For comparison, simple physical mixing of NiSn/MgO and NiSn@C catalysts were

also investigated for ethanol upgradation. Compared with the alone NiSn@C or NiSn/MgO catalyst, the catalyst exhibits better catalytic performance with ethanol conversion of 63.2 % and liquid products yield of 45.7 %, but obviously much lower than those of NiSn@C-MgO (Fig. 7a,b). NiSn/MgO has poor catalytic performance due to its poor hydrothermal stability [45,46]. NiSn@C floats on the liquid reactants due to its lipophilicity and low density, which is not conducive to the mass transfer between reaction substrate and catalyst. Therefore, we speculate that, on the one hand, the hydrothermal stability of the

Table 1Reusability of the spent NiSn/C-MgO catalyst for aqueous ethanol upgrading^a.

Cycle	C _{ethanol} (C-mol %)	Y _{liquid} (C-mol %)	Higher alcohols distribution (%) ^b		
			Butanol	C ₆₊ alcohols	Others
0	73.3	59.0	30.8	60.9	8.3
1	69.4	56.1	31.7	58.1	10.2
2	75.1	60.2	29.5	61.2	9.3
3	72.8	57.3	30.5	58.8	10.7
4	71.3	56.5	31.2	57.9	10.9

^aThe catalyst was recycled through centrifugation. Reaction conditions: 0.5 g of NaOH, 10 g of H₂O, 10 g ethanol, $T = 250\text{ }^{\circ}\text{C}$, $t = 12\text{ h}$. (C: conversion; Y: yield).

^bDistribution of higher alcohols in liquid products. Others: small amounts of aldehydes, trace amounts of alkanes, and other unidentified species.

NiSn/C-MgO catalyst is enhanced due to the protection of carbon shell, and on the other hand, the catalytic performance is enhanced due to its amphiphilicity.

To confirm the hydrophilicity/lipophilicity of NiSn/MgO, NiSn/C, and NiSn/C-MgO catalysts, the contact angle measurement of water and dodecanol on the surface of various catalysts was conducted. As shown in Fig. 8a, the water-droplet contact angle of NiSn/C is as high as 70.0° , indicating that NiSn/C had great lipophilicity. Obviously, more content of carbon would improve the lipophilicity of catalyst. The dodecanol-droplet contact angle is 24.1° (Fig. 8d), which further demonstrate that NiSn/C has great lipophilicity. Correspondingly, NiSn/MgO exhibits the good hydrophilicity, with a low water-droplet contact angle of 16.0° and a high dodecanol-droplet contact angle of 58.2° (Fig. 8b,e). Surely, NiSn/C-MgO, which has a moderate water-droplet contact angle of 54.0° and dodecanol-droplet contact angle of

44.9° (Fig. 8c,f), is both hydrophilic due to the hydrophilic groups such as hydroxyl groups on the surface of MgO and lipophilic due to the lipophilicity of carbon materials. To further investigate the effect of hydrophilicity and lipophilicity on the ethanol upgrading reaction, the properties of emulsion stabilized by different catalysts were investigated by a polarizing microscope. As can be seen from Fig. 8g,i, the average diameter of emulsion droplets stabilized by NiSn/MgO is $10.1\text{ }\mu\text{m}$, smaller than NiSn/C (average diameters of $25.1\text{ }\mu\text{m}$). Oxide nanoparticles (e.g MgO) are used to stabilize oil-in-water emulsions, due to their hydrophilicity, rough surface and asymmetric shape, which are beneficial for maintaining the stability of emulsion droplets [47]. Excitingly, the average diameter of emulsion droplets stabilized by NiSn/C-MgO is $14.1\text{ }\mu\text{m}$, which is similar to NiSn/MgO, implying that NiSn/C-MgO can function as an emulsifier to stabilize the emulsion. Furthermore, the images of the water phase and oil phase distributions of various catalysts in liquid products also visually showed the hydrophilicity and lipophilicity (Fig. 8j-l). In detail, NiSn/C and NiSn/MgO float on the top and sink to the bottom of liquid products, respectively, while NiSn/C-MgO is uniformly distributed in the liquid product.

To further confirm the significance of the regulation of the lipophilicity and hydrophilicity of NiSn/C-MgO on ethanol upgrading to higher alcohols, more characterizations were performed to exclude the influence of surface electronic properties on the catalytic performance. We collected the XPS spectra of NiSn/MgO, NiSn/C, and NiSn/C-MgO catalysts. As shown in Fig. S2, all catalysts display similar features in the corresponding XPS spectra. In particular, Ni mainly consists of Ni⁰ and Ni²⁺ in those catalysts, as revealed by the peaks at 852.8 and 853.7 eV (Table S1). The ratios of Ni⁰/Ni²⁺, which were obtained by fitting the XPS spectra, were close to 7/3 for various catalysts (Fig. S2a and Table S1). Moreover, results from XPS spectra indicate that Sn consists of

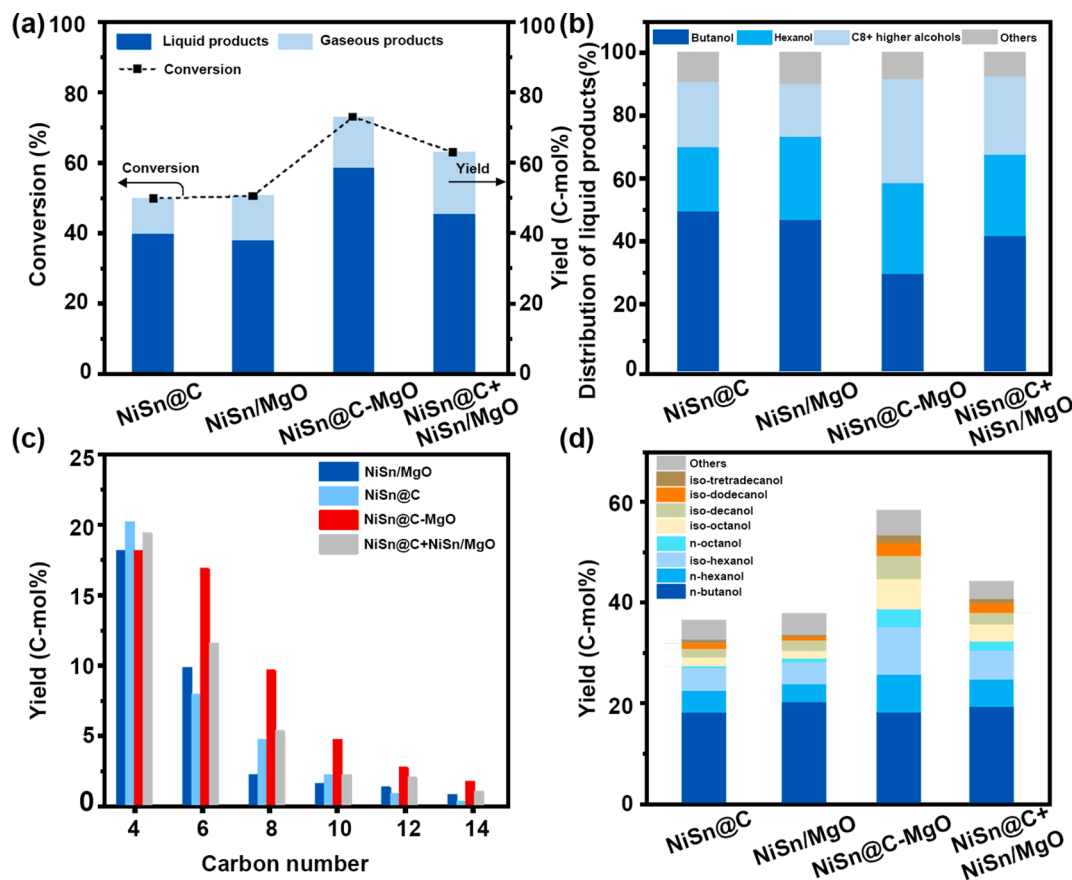


Fig. 7. A) ethanol conversion, yield of products; b) selectivity of liquid products; c) carbon number distributions of higher alcohols; d) yield of liquid products in ethanol upgrading reaction over various catalysts. reaction conditions: $m_{\text{NaOH}} = 0.5\text{ g}$, $m_{\text{catalyst}} = 0.5\text{ g}$ ($m_{\text{NiSn/MgO} + \text{NiSn/C}} = 0.25\text{ g}$ of NiSn/MgO + 0.25 g of NiSn/C), $m_{\text{ethanol}} = 10\text{ g}$, $m_{\text{water}} = 10\text{ g}$, $T = 250\text{ }^{\circ}\text{C}$, and $t = 12\text{ h}$.

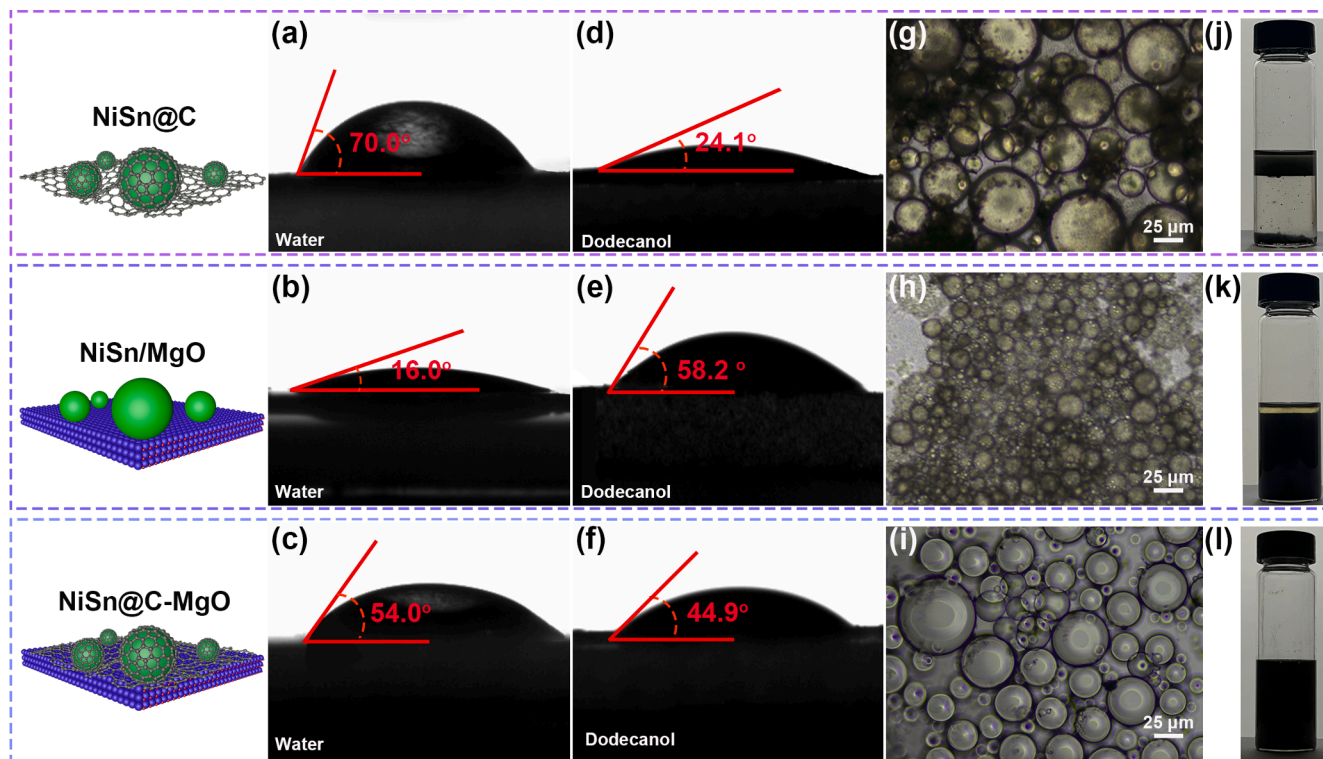


Fig. 8. The contact angle tests and emulsifying properties of NiSn@C, NiSn/MgO, NiSn@C-MgO. a-c) the contact angle tests image of the water droplet; d-e) the contact angle tests image of the dodecanol droplet; g-h) the optical microscope image; j-l) distributions of various catalysts in liquid product.

Sn^0 , Sn^{2+} , and Sn^{4+} in all catalysts. It is noted that only slight variations of $\text{Sn}^0/\text{Sn}^{2+}/\text{Sn}^{4+}$ molar ratios are observed in various catalysts (Fig. S2b and Table S2). The above results indicate that the current strategy for regulating the lipophilicity and hydrophilicity has negligible influence on the surface electronic properties of NiSn composite

nanoparticles. Consequently, we conclude that the enhanced performance for ethanol upgradation over NiSn@C-MgO is attributed to its amphipathic properties that can upgrade the ethanol in both oil and water phase.

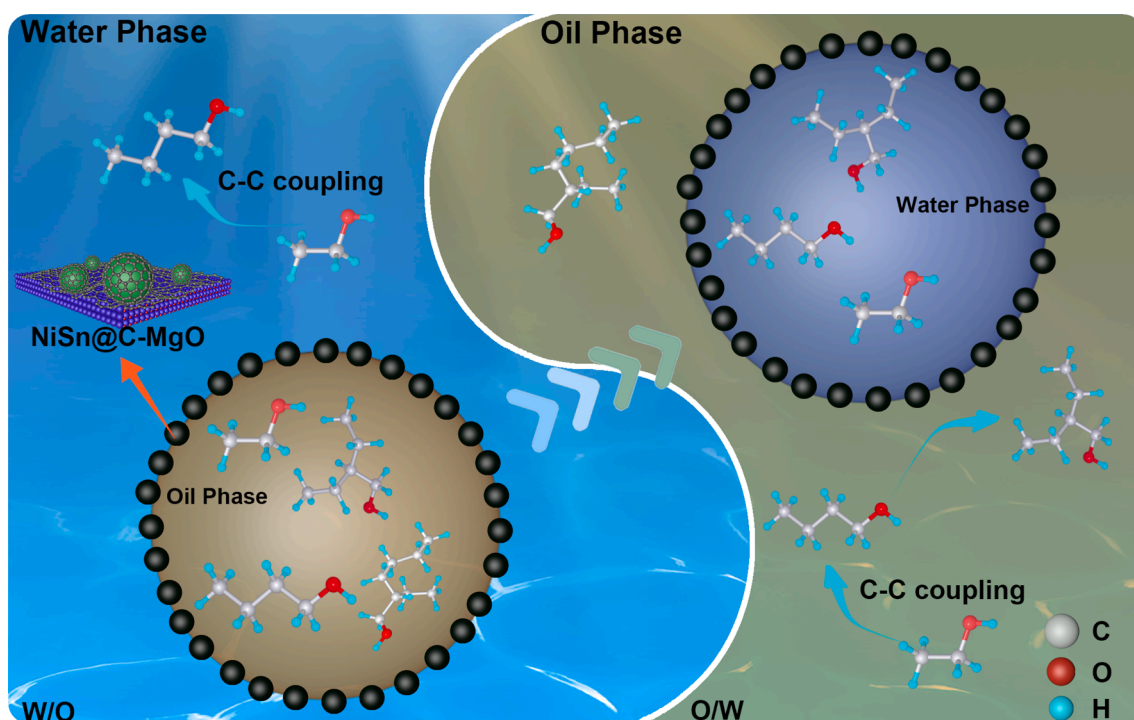


Fig. 9. Schematic illustration of the ethanol upgrading reaction pathway over NiSn@C-MgO catalysts. NiSn@C-MgO catalysts can stabilize oil–water emulsions and catalyze the reactions at the oil–water interface; the product migrates in both the oil and water phases as the reaction proceeds.

3.4. Reaction pathway of ethanol upgrading to higher alcohols

On the basis of the previous works [31,41,48], we can know that the process of aqueous ethanol upgrading through Guerbet route includes: firstly, the self-coupling of ethanol into butanol, and then the cross-coupling of ethanol and C_{4+} alcohols to generate C_{6+} alcohols. The oil–water interface gradually forms as the aqueous ethanol is converted to higher alcohols, and the liquid product subsequently separate into two independent phases, which is ascribed to the hydrophobic properties of C_{4+} higher alcohols. Since traditional heterogeneous catalysts can only exist stably in a single phase, the cross-coupling of ethanol and butanol (or C_{6+} alcohols) will be limited. In view of this, we proposed a possible reaction pathway for aqueous ethanol upgrading over NiSn@C-MgO catalyst, in which NiSn NPs are used as metal active center and MgO act as the hydrophilic support, while carbon shells as lipophilic support protect NiSn nanoparticles from agglomeration. As shown in Fig. 9, at the initial stage of the reaction, a single aqueous phase is present due to the mutual solubility of ethanol and water. With the gradual generation of C_{4+} alcohols, an oil-in-water emulsion system is formed, owing to the stability of emulsion droplets by NiSn@C-MgO. In the emulsion, it is favorable for the contact between water-soluble substrates (C_2 – C_4 , etc.) and oil-soluble products (C_{4+} , etc.), and strengthens the cross-coupling reaction process, which will get full use of the reactants (ethanol, *n*-butanol, etc.) [29]. Ultimately, through the emulsion catalytic effect, the generated *n*-butanol further convert into C_{4+} higher alcohols.

4. Conclusion

In conclusion, an amphiphilic nanohybrid composed of NiSn NPs coated with carbon shell and MgO nanoflakes supports were synthesized via a simple one-step carbonization process. The NiSn@C-MgO catalyst can stabilize emulsions and facilitate the cross-coupling of ethanol and C_{4+} alcohols to generate C_{6+} alcohols in aqueous ethanol upgrading. The carbonization temperature has a great influence on the structure of NiSn@C-MgO, which further greatly affects the catalyst performance. The optimal NiSn@C-MgO-550 displays an excellent activity with ethanol conversion of 73.3 %, the C_{6+} alcohols and C_{8+} alcohols selectivity in liquid products also reach up to 60.9 and 32.5 % (Yield C_{6+} alcohols = 36.0 % and Yield C_{8+} alcohols = 19.2 %), respectively. Furthermore, tuning the hydrophilic and lipophilic properties of catalysts play a key role in catalytic activity during aqueous ethanol upgrading to higher alcohols. This present strategy may not only provide an efficient strategy for the direct upgrading of water-containing bioethanol to desired biofuel components, but also have great potential in many important applications such as catalysis, emulsifiers and design of functional nanomaterials.

CRedit authorship contribution statement

Xueying Cai: Conceptualization, Methodology, Validation, Investigation, Writing – original draft. **Xiyin Li:** Investigation. **Minyang Dang:** Data curation. **Huanxin Huang:** Investigation, Data curation. **Ailin Liu:** Formal analysis. **Qian Zhang:** Supervision, Resources. **Yunhong Pi:** Supervision, Formal analysis. **Qingwei Meng:** Supervision, Methodology. **Xiaoping Wu:** Conceptualization, Methodology, Writing – review & editing. **Tiejun Wang:** Methodology, Resources, Funding acquisition, Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (U21A20288, 52106232, 51876046), the National Key Research and Development Project (2018YFE0125200), and the Guangdong Basic and Applied Basic Research Foundation (2021A1515010576). They are especially grateful to Liang Zheng (Guangzhou Institute of Energy Conversion, Chinese Academy of Sciences) for TEM technique.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2022.126971>.

References

- [1] He Y, Zhu L, Luo Y, Fan M, Yang M, Zhang Y, et al. Selective upgrading of biomass pyrolysis oil into renewable p-xylene with multifunctional M/SiO₂/HZSM-5 catalyst. *Fuel Process Technol* 2021;213:106674.
- [2] Zhao J, Xu Y, Wang W, Griffin J, Roozeboom K, Wang D. Bioconversion of industrial hemp biomass for bioethanol production: A review. *Fuel* 2020;281:118725.
- [3] He YT, Luo YH, Yang MY, Zhang YH, Fan MH, Li QX. High value utilization of biomass: selective catalytic transformation of lignocellulose to bio-based 2,5-dimethylphenol. *Catal Sci Technol* 2022;12:4524–34.
- [4] Wang H, Miao G, Kong L, Luo H, Zhang Y, Zhao X, et al. Efficient one-pot valorization of ethanol to 1-butanol over an earth-abundant Ni-MgO catalyst under mild conditions. *Sustain Energy Fuels* 2020;4(4):1612–5.
- [5] Pang JF, Zheng MY, Wang ZN, Liu SM, Li XS, Li XQ, et al. Catalytic upgrading of ethanol to butanol over a binary catalytic system of FeNiOx and LiOH. *Chinese J Catal* 2020;41:672–768.
- [6] Aitchison H, Wingard RL, Wass DF. Homogeneous ethanol to butanol catalysis-Guerbet renewed. *ACS Catal* 2016;6(10):7125–32.
- [7] Yuan B, Zhang J, An Z, Zhu Y, Shu X, Song H, et al. Atomic Ru catalysis for ethanol coupling to C_{4+} alcohols. *Appl Catal B-Environ* 2022;309:121271.
- [8] Jordison TL, Lira CT, Miller DJ. Condensed-phase ethanol conversion to higher alcohols. *Ind Eng Chem Res* 2015;54(44):10991–1000.
- [9] Nezam I, Zak J, Miller DJ. Condensed-phase ethanol conversion to higher alcohols over bimetallic catalysts. *Ind Eng Chem Res* 2020;59(31):13906–15.
- [10] Rorrer J, Pindi S, Toste FD, Bell AT. Effect of alcohol structure on the kinetics of etherification and dehydration over tungstated zirconia. *ChemSusChem* 2018;11(18):3104–11.
- [11] Li H, Gui Z, Yang S, Qi Z, Saravanamurugan S, Riisager A. Catalytic tandem reaction for the production of jet and diesel fuel range alkanes. *Energy Technol-Ger* 2018;6(6):1060–6.
- [12] Jeong H, Bathula HB, Kim TW, Han GB, Jang JH, Jeong B, et al. Superior long-term stability of a mesoporous alumina-supported Pt catalyst in the hydrodeoxygenation of palm oil. *ACS Sustain Chem Eng* 2021;9(3):1193–202.
- [13] Panpian P, Tran TTV, Kongparakul S, Attanatho L, Thanmongkhon Y, Wang P, et al. Production of bio-jet fuel through ethylene oligomerization using NiAlKIT-6 as a highly efficient catalyst. *Fuel* 2021;287:119831.
- [14] Babu R, Subaramanian M, Midya SP, Balaraman E. Nickel-catalyzed Guerbet type reaction: C-alkylation of secondary alcohols via double (de)hydrogenation. *Org Lett* 2021;23(9):3320–5.
- [15] Wang K, Zhang L, Tang W, Sun H, Xue D, Lei M, et al. Asymmetric Guerbet reaction to access chiral alcohols. *Angew Chem Int Ed* 2020;59(28):11408–15.
- [16] Tramontina R, Galman JL, Parmeggiani F, Derrington SR, Bugg TDH, Turner NJ, et al. Consolidated production of coniferol and other high-value aromatic alcohols directly from lignocellulosic biomass. *Green Chem* 2020;22(1):144–52.
- [17] Kondo M, Nakamura K, Krishnan CG, Takizawa S, Abe T, Sasai H. Photoswitchable chiral phase transfer catalyst. *ACS Catal* 2021;11(3):1863–2187.
- [18] Onkarappa SB, Dutta S. Phase transfer catalyst assisted one-pot synthesis of 5-(chloromethyl)furfural from biomass-derived carbohydrates in a biphasic batch reactor. *ChemistrySelect* 2019;4(25):7502–6.
- [19] Rull J, Jara JJ, Sebastián RM, Vallribera A, Nájera C, Majoral J-P, et al. Recoverable dendritic phase-transfer catalysts that contain(+)-cinchonine-derived ammonium salts. *ChemCatChem* 2016;8(12):2049–56.
- [20] Hua M-Q, Cui H-F, Wang L, Nie J, Ma J-A. Reversal of enantioselectivity by tuning the conformational flexibility of phase-transfer catalysts. *Angew Chem Int Ed* 2010;49(15):2772–6.
- [21] Golandaj A, Ahmad A, Ramjugernath D. Phosphonium salts in asymmetric catalysis: a journey in a decade's extensive research work. *Adv Synth Catal* 2017;359(21):3676–706.

- [22] He R, Ding C, Maruoka K. Phosphonium salts as chiral phase-transfer catalysts: asymmetric Michael and Mannich reactions of 3-aryloxindoles. *Angew Chem Int Ed* 2009;48(25):4559–61.
- [23] Shirakawa S, Maruoka K. Recent developments in asymmetric phase-transfer reactions. *Angew Chem Int Ed* 2013;52(16):4312–48.
- [24] Kirschner S, Peters M, Yuan K, Uzelac M, Ingleson MJ. Developing organoboranes as phase transfer catalysts for nucleophilic fluorination using CsF. *Chem Sci* 2022; 13(9):2661–8.
- [25] Zhou Yu, Guo Z, Hou W, Wang Q, Wang J. Polyoxometalate-based phase transfer catalysis for liquid–solid organic reactions: a review. *Catal Sci Technol* 2015;5(9): 4324–35.
- [26] Xie X, Brown JS, Joseph PJ, Liotta CL, Eckert CA. Phase-transfer catalyst separation by CO₂ enhanced aqueous extraction. *Chem Commun* 2002;(10):1156–7.
- [27] Ruiz MP, Faria J, Shen M, Drexler S, Prasomsri T, Resasco DE. Nanostructured carbon-metal oxide hybrids as amphiphilic emulsion catalysts. *ChemSusChem* 2011;4(7):964–74.
- [28] Zhou X, Deng J, Yang R, Zhou D, Fang C, He X, et al. Facile preparation and characterization of fibrous carbon nanomaterial from waste polyethylene terephthalate. *Waste Manag* 2020;107:172–81.
- [29] Crossley S, Faria J, Shen M, Resasco DE. Solid nanoparticles that catalyze biofuel upgrade reactions at the water/oil interface. *Science* 2010;327(5961):68–72.
- [30] Shan Y, Yu C, Yang J, Dong Q, Fan X, Qiu J. Thermodynamically stable pickering emulsion configured with carbon-nanotube-bridged nanosheet-shaped layered double hydroxide for selective oxidation of benzyl alcohol. *ACS Appl Mater Inter* 2011;4(22):12203–9.
- [31] Wu X, Cai X, Zhang Q, Bi P, Meng Q, Pi Y, et al. Upgrading of aqueous bioethanol to higher alcohols over NiSn/MgAlO catalyst. *ACS Sustain Chem Eng* 2021;9(33): 11269–79.
- [32] Sui X-L, Wang Z-B, Li C-Z, Zhang J-J, Zhao L, Gu D-M. Effect of core/shell structured TiO₂@C nanowire support on the Pt catalytic performance for methanol electrooxidation. *Catal Sci Technol* 2016;6(11):3767–75.
- [33] Liu Q, Wang H, Xin H, Wang C, Yan L, Wang Y, et al. Selective cellulose hydrogenolysis to ethanol using Ni@C combined with phosphoric acid catalysts. *ChemSusChem* 2019;12(17):3977–87.
- [34] Ni D, Wang L, Sun Y, Guan Z, Yang S, Zhou K. Amphiphilic hollow carbonaceous microspheres with permeable shells. *Angew Chem Int Ed* 2010;49(25):4223–7.
- [35] Fang S, Cui Z, Zhu Y, Wang C, Bai J, Zhang X, et al. In situ synthesis of biomass-derived Ni/C catalyst by self-reduction for the hydrogenation of levulinic acid to γ -valerolactone. *J Energy Chem* 2019;37:204–14.
- [36] Sharma A, Saito I, Nakagawa H, Miura K. Effect of carbonization temperature on the nickel crystallite size of a Ni/C catalyst for catalytic hydrothermal gasification of organic compounds. *Fuel* 2007;86(7–8):915–20.
- [37] Xie G-Q, Lu J-Q, Zheng H-Y, Liu X-J, Luo M-F, Li X-N. Effect of carbonization temperature on the textural properties of Ce_{0.8}Zr_{0.2}O₂ solid solution by an improved citrate sol–gel method. *J Alloy Compd* 2010;493(1–2):169–74.
- [38] Zhang P, Yu C, Fan X, Wang X, Ling Z, Wang Z, et al. Magnetically recoverable Ni/C catalysts with hierarchical structure and high-stability for selective hydrogenation of nitroarenes. *Phys Chem Chem Phys* 2015;17(1):145–50.
- [39] Zhu C, Xin H, Hong J, Wang H, Wang C, Ma L, et al. Hydrocarbon distribution of cellulose hydrogenolysis over Ru–MoO_x/C combined with HZSM-5. *ACS Sustain Chem Eng* 2021;9(44):14704–21.
- [40] Petraki F, Papaefthimiou V, Kennou S. A study of the Ni-phthalocyanine/gold interface using x-ray and ultraviolet photoelectron spectroscopies. *J Phys Conf Ser* 2005;10:135–8.
- [41] Liu W, Chen Bo, Zhang Q, Qiu S, Wu X, Meng Q, et al. One-pot synthesis of high-carbon bio-alcohols from aqueous ethanol upgrading over water-tolerance NiSn@C catalyst. *Energy Convers Manage* 2021;249:114822.
- [42] Wang Q, Yang R, Li J, Yang Y, Wu Y, Zhou X, et al. 2D DUT-8(Ni)-derived Ni@C nanosheets for efficient hydrogen evolution. *J Solid State Electrochem* 2020;24 (10):2461–7.
- [43] Ma Y, Chen M, Geng H, Dong H, Wu P, Li X, et al. Synergistically tuning electronic structure of porous β -Mo₂C spheres by Co doping and Mo-vacancies defect engineering for optimizing hydrogen evolution reaction activity. *Adv Funct Mater* 2020;30(19):2000561.
- [44] Baggetto L, Ganesh P, Meisner RP, Unocic RR, Jumas J-C, Bridges CA, et al. Characterization of sodium ion electrochemical reaction with tin anodes: Experiment and theory. *J Power Sources* 2013;234:48–59.
- [45] Zhu Q-C, Shen B-X, Ling H, Gu R. Effect of various supports on the catalytic activity of Pd for cumene hydroperoxide hydrogenation. *Catal Lett* 2009;132(3–4):464–71.
- [46] Li S, Guo L, Zhu C, Lu Y. Co-precipitated Ni–Mg–Al catalysts for hydrogen production by supercritical water gasification of glucose. *Int J Hydrogen Energy* 2013;38(23):9688–700.
- [47] Yu C, Fan L, Yang J, Shan Y, Qiu J. Phase-reversal emulsion catalysis with CNT-TiO₂ nanohybrids for the selective oxidation of benzyl alcohol. *Chem-Eur J* 2013; 19(48):16192–5.
- [48] Zhang Q, Liu W, Chen B, Qiu S, Wang T. Upgrading of aqueous ethanol to fuel grade higher alcohols over dandelion-like Ni-Sn catalyst. *Energy Convers Manage* 2020;216:112914.